

STA 2101: Statistics & Probability

Lecture 01: Introduction to the Course

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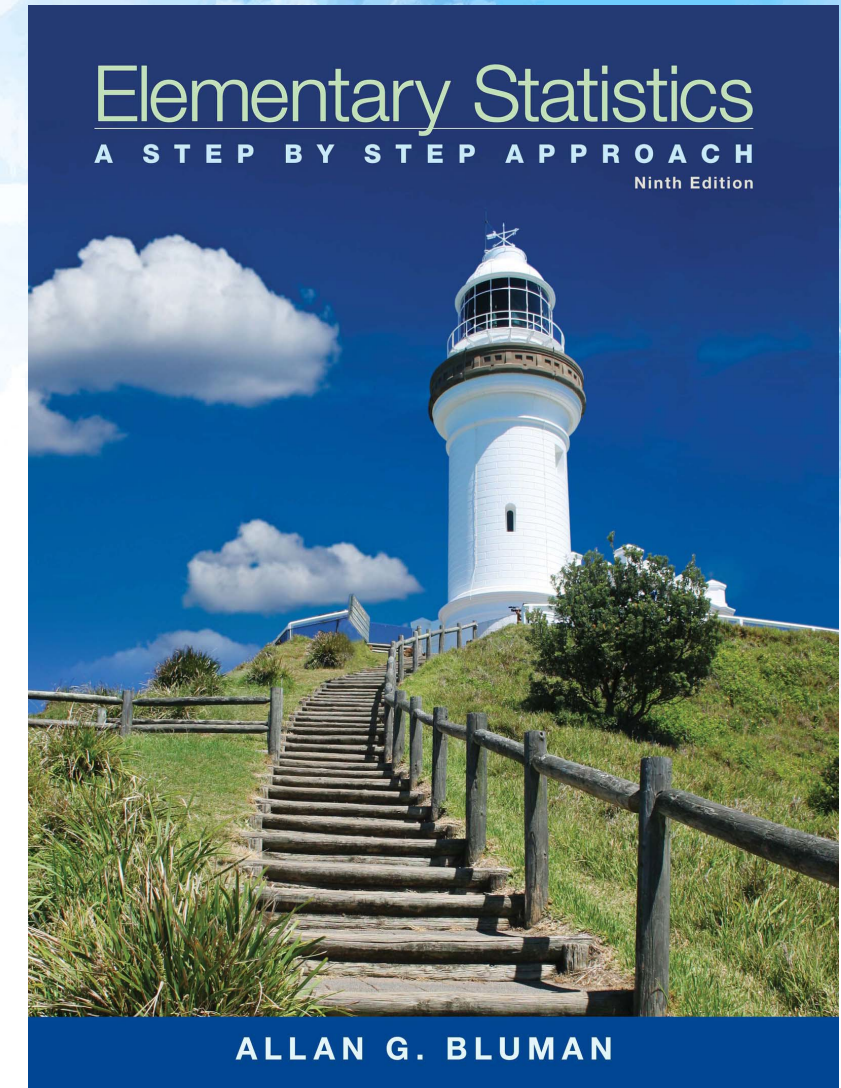
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Fall 2025

Reference Book

- **Text Book(s):**
 - Elementary Statistics – Allan G. Bluman
 - Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver
- **Reference Material/Book(s):**
 - S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



The Nature of Probability and Statistics

Statistics is the science of conducting studies to collect, organize, summarize, analyze, and draw conclusions from data.

Statistics is the theory and method of analyzing quantitative data obtained from samples of observations in order to study and compare sources of variance of phenomena, to **help make decisions** to accept or reject hypothesized relations between the phenomena, and to aid in **drawing reliable inferences** from empirical observations.

The Nature of Probability and Statistics

The FBI reported that violent crimes were down by 6.4% in 2011.

- USA TODAY reported that the average college graduate student loan debt was about \$19,000.
- The College Stress and Mental Illness Poll reported that 85% of college and university students reported feeling stress daily; 77% reported stress from school work, and 74% experienced stress from grades.
- The Occupational Outlook Handbook reported that the median hourly wage for a registered nurse was \$31.10 per hour.
- Reader's Digest reported that the average cost of using a plasma television is \$0.1152 per hour.
- In 2013, the number of sales of smartphones is estimated to be 832.5 million units globally.

Statistics is used in almost all fields of human endeavor. In sports, for example, a statistician may keep records of the number of yards a running back gains during a football game, or the number of hits a baseball player gets in a season. In other areas, such as public health, an administrator might be concerned with the number of residents who contract a new strain of flu virus during a certain year. In education, a researcher might want to know if new methods of teaching are better than old ones. These are only a few examples of how statistics can be used in various occupations.

In Short

- **Definition:** Statistics is the science of collecting, organizing, analyzing, and interpreting data to make decisions.
- **Purpose:** Helps transform raw numbers into meaningful information.
- **Example:**
 - A school collects test scores of 500 students.
 - Statistics helps summarize average marks, distribution of scores, and predict future performance.

Application of statistics

- The field of statistics is the science of learning from data.
- It is a tool we use to convert data into information.
- Applications of statistics is very wide. Different and multidisciplinary fields evolved over time.

Visualization



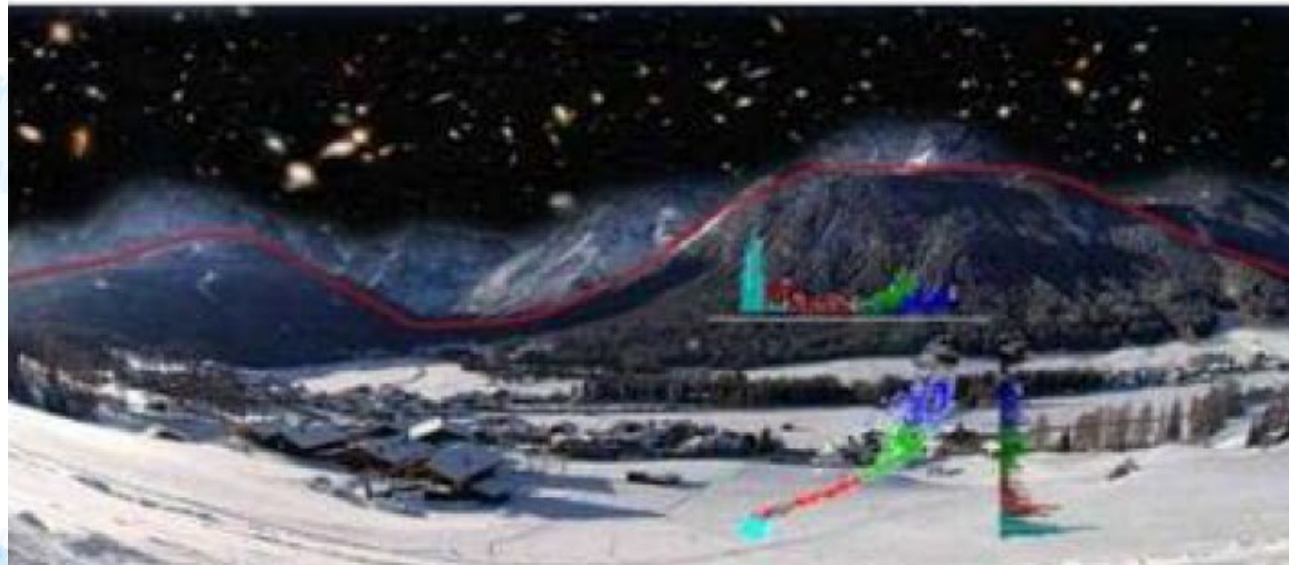
- **Data Mining:** Finding useful patterns in data.
- **Data-Driven Decisions:** Making informed choices based on data analysis.
- **Data Summarization:** Visualizing and summarizing data with charts and graphs.
- **Databases:** Optimizing database performance and storage.
- **Artificial Intelligence:** Validating and improving AI models.
- **Bioinformatics:** Analyzing biological data.
- **Natural Language Processing (NLP):** Analyzing and understanding text.
- **Computer Vision:** Processing and understanding images.

Applications in CSE

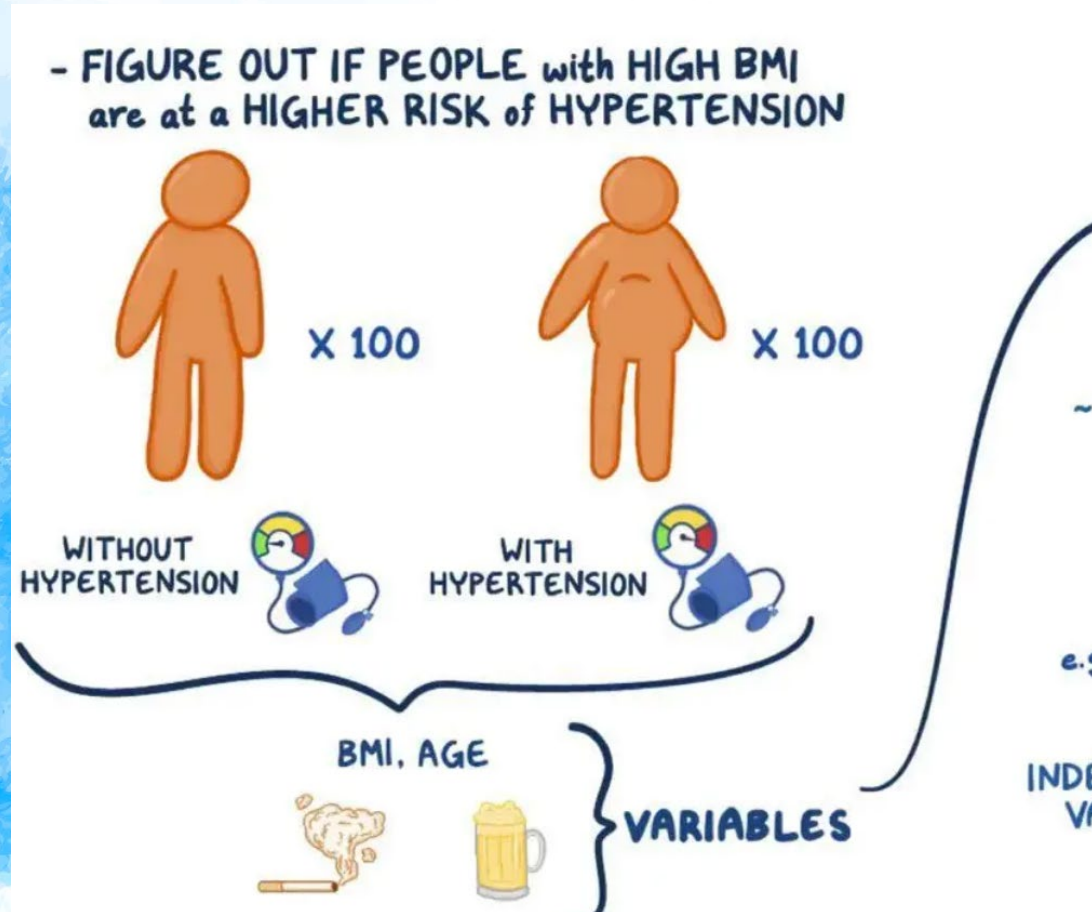
- **Data Mining & Data-Driven Decisions:** Statistics helps find useful patterns in data, enabling businesses to make informed, data-driven decisions. Methods like regression and clustering are key to this process.
- **Data Summarization:** Statistical concepts are essential for summarizing and visualizing data through charts, graphs, and dashboards. The **Pearson Correlation Coefficient**, for example, measures the relationship between variables.
- **Databases:** Statistical models are used to analyze database performance, detect outliers, and optimize data storage using techniques like **Chi-Square Tests** and correlation analysis.
- **Artificial Intelligence (AI) & Neural Networks:** Statistical tests are used for AI model validation and performance analysis. Concepts like cross-validation, regularization, and gradient-based optimization are fundamental to building and evaluating machine learning models.
- **Bioinformatics:** Statistics is crucial for analyzing biological data, including identifying disease markers and predicting protein structures.
- **Natural Language Processing (NLP) & Computer Vision:** Statistical models are widely used in NLP for tasks like text prediction and information retrieval. In computer vision, they are vital for image processing tasks such as feature extraction and dimensionality reduction using techniques like **Principal Component Analysis (PCA)** and **Singular Value Decomposition (SVD)**.

Application of Statistics to other disciplines

- Astrostatistics - The field which applies statistics to astronomical data
- The combination of astrophysics and statistical analysis.

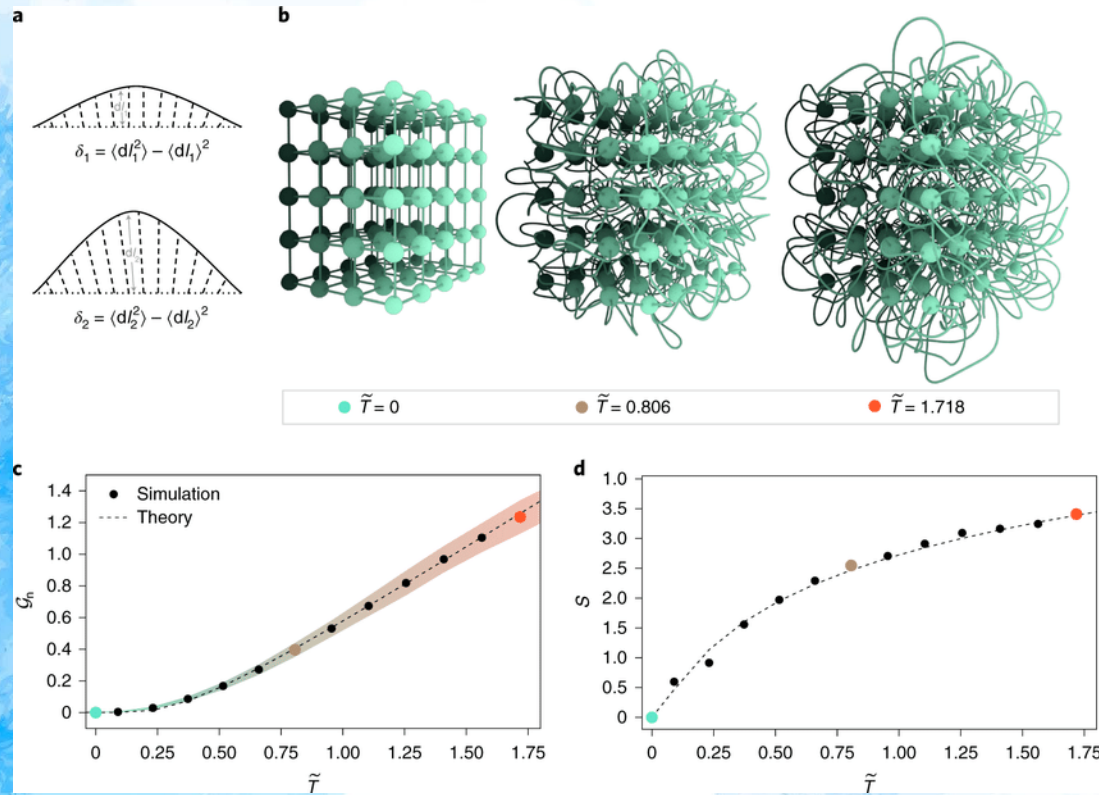


Application of Statistics to other disciplines



- Biostatistics - Biostatistics is a scientific discipline that uses statistical theory and methods to address problems in biology, medicine, and public health

Application of Statistics to other disciplines



- Statistical physics - It is a branch of physics that applies probability theory and statistics to understand the behavior of macroscopic systems composed of a large number of microscopic entities, such as atoms or molecules

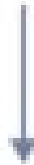
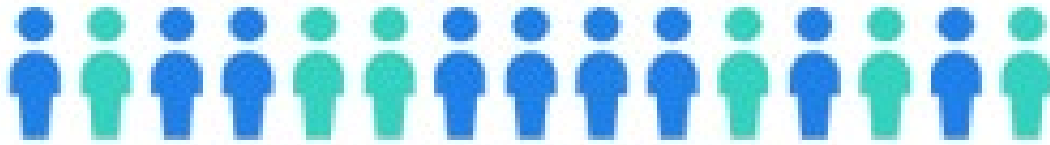
The Foundations of Statistics

- **Variables and Data**

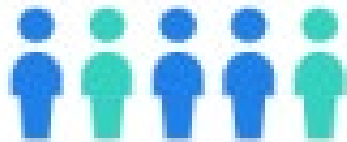
- In statistics, we use **variables** to describe and understand seemingly random situations. A variable is a characteristic or attribute that can take on different values. The actual measurements or observations a variable can assume are called **data**. A collection of these data values is known as a **data set**, and each individual value within it is a **data value** or **datum**. When a variable's values are determined by chance, it's called a **random variable**.
- For example, an insurance company might analyze historical records to understand accident rates. They find that on average, 3 out of every 100 cars they insure are involved in an accident each year. While they can't predict which specific cars will crash, they can use this **general pattern** to set their insurance rates. This is a classic example of a random variable—the specific outcome (which car has an accident) is random, but the overall probability (3%) is stable and predictable over the long term.

The Foundations of Statistics

Population

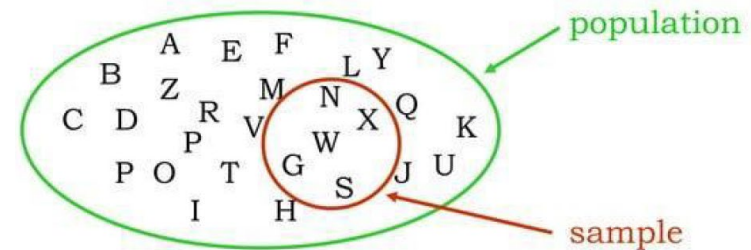


Sample



Variable: A variable is a characteristic or attribute that can assume different values. Data are the values (measurements or observations) that the variables can assume.

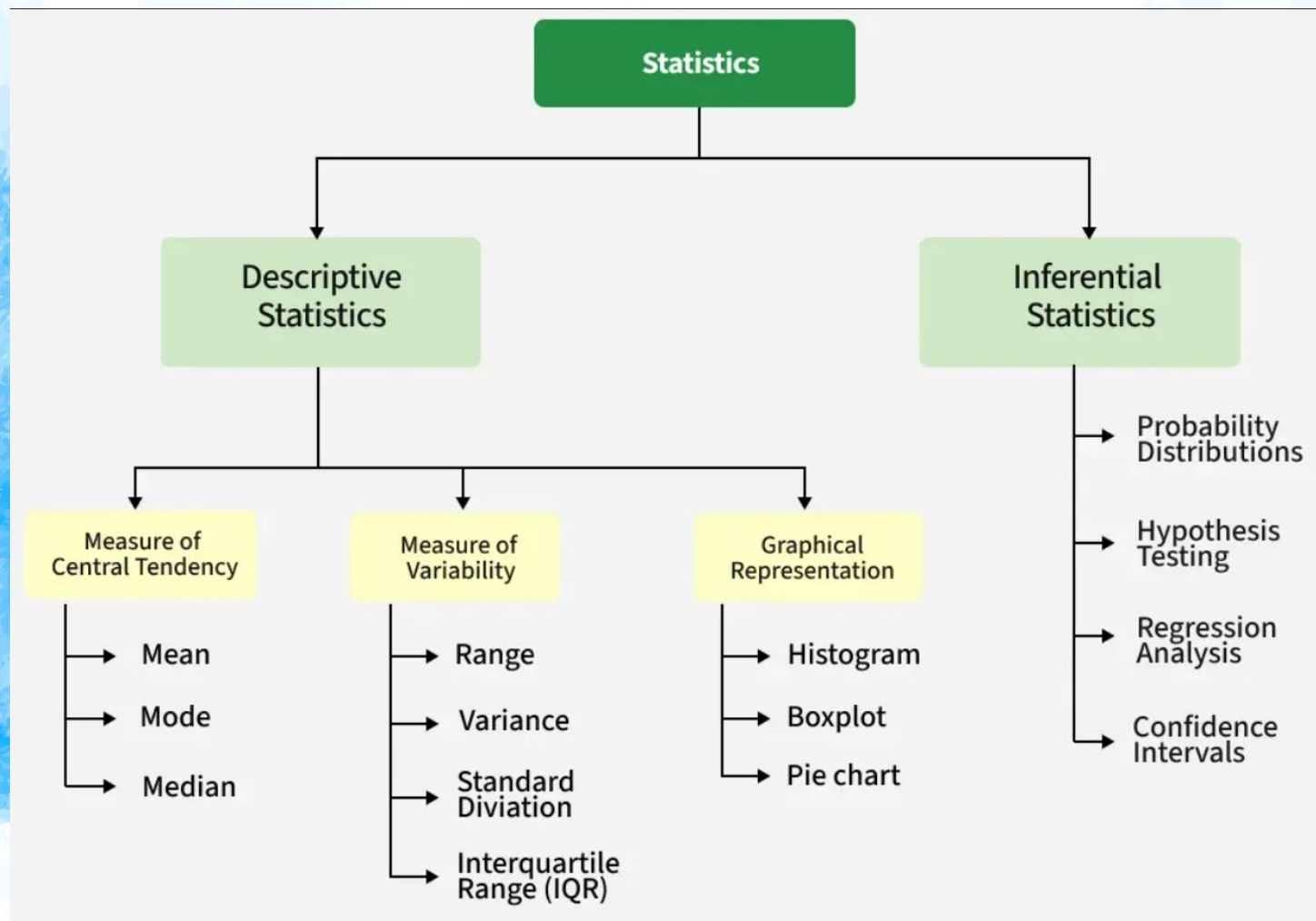
Population and sample: A **population** is the entire group that you want to draw conclusions about. A **sample** is the specific group that you will collect data from. The size of the sample is always less than the total size of the population.



Population vs Sample

- A **population** includes every subject (human or otherwise) being studied. When data is collected from every subject in a population, the process is called a **census**. A well-known example is the **U.S. Census**, which is conducted every 10 years to determine the apportionment of seats in the House of Representatives.
- A **sample** is a subset of the population. Since it is often too expensive, time-consuming, or impractical to collect data from an entire population, researchers typically use a sample to draw conclusions about the whole.

Types of Statistics



- Statistics is sometimes **divided into two main areas**, depending on how data are used. The two areas are:
- 1. Descriptive statistics
- 2. Inferential statistics

Descriptive Statistics

- **1. Descriptive statistics:** Descriptive statistics consists of the collection, organization, summarization, and presentation of data. In descriptive statistics the statistician tries to describe a situation.
- Example: Consider the national census conducted by the U.S. government every 10 years. Results of this census give you the average age, income, and other characteristics of the U.S. population. To obtain this information, the Census Bureau must have some means to collect relevant data.
- Once data are collected, the bureau must organize and summarize them. Finally, the bureau needs a means of presenting the data in some meaningful form, such as charts, graphs, or tables.

Inferential statistics

- **2. Inferential statistics:** Inferential statistics consists of generalizing from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions.
- Unlike descriptive statistics, which simply summarizes known data, inferential statistics makes inferences or draws conclusions that go beyond the available data. It uses probability theory to estimate population parameters and test hypotheses. By working with a sample, inferential statistics allows researchers to make informed decisions without having to gather data from an entire population.

In Short

- **Descriptive Statistics:** Summarizes and describes data.
- Example: The average height of 100 students is **165 cm**.
- **Inferential Statistics:** Uses a sample to draw conclusions about a larger population.
- Example: Surveying 200 voters to predict the outcome of a national election.

Difference Between the two

Descriptive Statistics	Inferential Statistics
It gives information about raw data which describes the data in some manner.	It makes inferences about the population using data drawn from the population.
It helps in organizing, analyzing, and to present data in a meaningful manner.	It allows us to compare data, and make hypotheses and predictions.
It is used to describe a situation.	It is used to explain the chance of occurrence of an event.
It explains already known data and is limited to a sample or population having a small size.	It attempts to reach the conclusion about the population.
Examples include: mean, median, mode, range, variance, histograms, pie charts.	Examples include: confidence intervals, hypothesis testing, regression models, p-values.
Limited to presenting and analyzing known data.	Allows predictions and conclusions that go beyond the data at hand.
Used for describing trends, organizing data for presentation.	Used for predicting trends, testing hypotheses, generalizing data from sample to population.
It can be achieved with the help of charts, graphs, tables, etc.	It can be achieved by <u>probability</u> .

Objective of Inferential Statistics

- **Why we use it:** It's not possible to study the entire population (too large, time-consuming, costly).
- **Objective:**
 1. Draw conclusions about population based on sample data.
 2. Measure the reliability of those conclusions.
- **Example:**
 - A company tests **50 bulbs** out of **10,000** produced.
 - If 47 bulbs work fine, they infer that ~94% of all bulbs are good.

Exercise from the book

EXAMPLE 1-1 Descriptive or Inferential Statistics

Determine whether descriptive or inferential statistics were used.

- a.* The average jackpot for the top five lottery winners was \$367.6 million.
- b.* A study done by the American Academy of Neurology suggests that older people who had a high caloric diet more than doubled their risk of memory loss.
- c.* Based on a survey of 9317 consumers done by the National Retail Federation, the average amount that consumers spent on Valentine's Day in 2011 was \$116.
- d.* Scientists at the University of Oxford in England found that a good laugh significantly raises a person's pain level tolerance.

SOLUTION

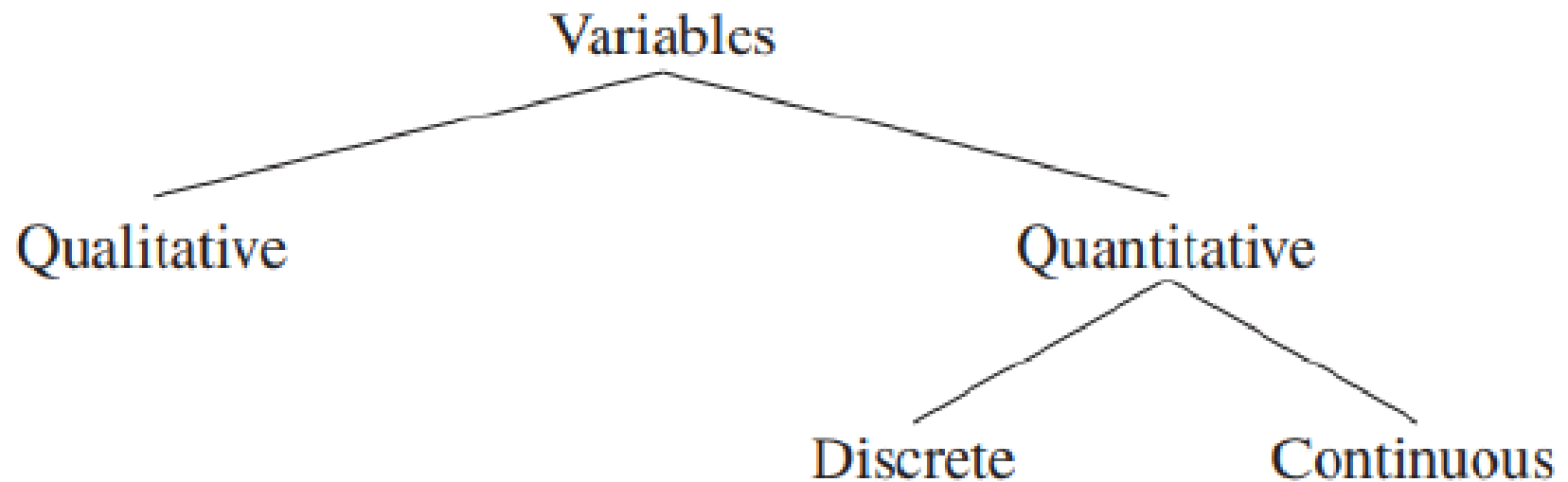
- a.* Descriptive statistics were used because this is an average, and it is based on data obtained from the top five lottery winners at this time.
- b.* Inferential statistics were used since this is a generalization made from a sample to a population.
- c.* Descriptive statistics were used since this is an average based on a sample of 9317 respondents.
- d.* Inferential statistics were used since an inference is made from a sample to a population.

Variables and Data Types

- Variables are characteristics or attributes that can be classified into two main types: **qualitative** and **quantitative**.
- **Qualitative Variables**
- **Qualitative variables** represent categories or descriptions and cannot be counted or measured numerically.
- **Example:** Gender (male, female, other) or hair color (black, brown, blonde).
- **Quantitative Variables**
- **Quantitative variables** are numerical and can be counted or measured.
- **Example:** Age, height, or weight.
- Quantitative variables are further divided into two subgroups:
- **Discrete Variables:** These variables can be counted and assume specific, distinct values. They typically represent whole numbers.
 - **Example:** The number of students in a class or the number of cars in a parking lot.
- **Continuous Variables:** These variables can assume any value within a given range, including fractions and decimals. They are obtained by measuring.
 - **Example:** A person's height (e.g., 175.5 cm) or the temperature of a room (e.g., 22.8°C)

Variables and Data Types

The classification of variables can be summarized as follows:



Example from book

EXAMPLE 1–2 Discrete or Continuous Variables

Classify each variable as a discrete variable or a continuous variable.

- a.* The highest wind speed of a hurricane
- b.* The weight of baggage on an airplane
- c.* The number of pages in a statistics book
- d.* The amount of money a person spends per year for online purchases

SOLUTION

- a.* Continuous, since wind speed must be measured
- b.* Continuous, since weight is measured
- c.* Discrete, since the number of pages is countable
- d.* Discrete, since the smallest value that money can assume is in cents

Class Boundaries

- Class boundaries represent the range of values that a rounded data point could fall within before rounding.
- For example, a recorded height of 73 inches could actually be anywhere between 72.5 and 73.5 inches. Boundaries are written for convenience, like 72.5–73.5, but mean all values up to, but not including, the upper limit.

TABLE 1–1 Recorded Values and Boundaries		
Variable	Recorded value	Boundaries
Length	15 centimeters (cm)	14.5–15.5 cm
Temperature	86 degrees Fahrenheit (°F)	85.5–86.5°F
Time	0.43 second (sec)	0.425–0.435 sec
Mass	1.6 grams (g)	1.55–1.65 g

Exercise from Book

EXAMPLE 1–3 Class Boundaries

Find the boundaries of each variable.

- a. 8.4 quarts
- b. 138 mmHg
- c. 137.63 mg/dL

SOLUTION

- a. 8.35–8.45 quarts
- b. 137.5–138.5 mmHg
- c. 137.625–137.635 mg/dL

- Class boundaries represent the range of values that a rounded data point could fall within before rounding.
- For example, a recorded height of 73 inches could actually be anywhere between 72.5 and 73.5 inches. Boundaries are written for convenience, like 72.5–73.5, but mean all values up to, but not including, the upper limit.



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Thank You

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